

MACHINE LEARNING IN PHYSICS SIMULATION-BASED INFERENCE

HARRISON B. PROSPER

PHY6937 / PHY4936

INTRODUCTION

Introduction: Statistical Inference

Consider the following recent *measurement* of the Hubble constant¹

$$H_0 = 73.30 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

What is the statistical meaning of the above statement?

If this a **Bayesian** statement, the meaning is: There is a 0.68 probability that

$$H_0 \in (72.36, 74.44)$$

1. Adam G. Riess *et al.* 2022. A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km s⁻¹ Mpc⁻¹ Uncertainty from the Hubble Space Telescope and the SH0ES Team. 2022 *ApJL* **934** L7.

Introduction: Statistical Inference

If, however,

$$H_0 = 73.30 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

is a **frequentist** statement, the meaning is more involved.

The meaning is that $H_0 \in (72.36, 74.44)$ is a member of an ensemble of statements 68% of which are true.

$(72.36, 74.44)$ is an example of a **confidence interval**. If the observations are repeated (infinitely!) many times, it is *guaranteed* that *at least* 68% of the intervals obtained will include the true value of the Hubble constant, *whatever the true value might be*.

Introduction: Statistical Inference

The key quantity in any serious statistical inference is the statistical model

$$p(X | \theta)$$

where X denotes the potential observations and θ the parameters of the statistical model, e.g., the Hubble constant.

When observations D are entered into the statistical model, the latter becomes the **likelihood function**

$$p(D | \theta)$$

which is key for constructing confidence intervals and, more generally, **confidence sets**.

Introduction: Statistical Inference

- Unfortunately, statistical models can be intractable.
- However, if one has access to a high-fidelity simulator, it is possible to make inferences without *explicit* knowledge of the statistical model.
- **Simulation-based inference** (SBI)^{1,2,3} are methods for making inferences from real data using a simulator.

1. Kyle Cranmer, Johann Brehmer, and Gille Loupes, <https://doi.org/10.1073/pnas.1912789117>;
2. Likelihood-Free Frequentist Inference: Confidence Sets with Correct Conditional Coverage Niccolo Dalmaso, Luca Masserano, David Zhao, Rafael Izbicki, Ann B. Lee, arXiv:2107.03920v6 [stat.ML] 6 Apr 2023.
3. A. Al Kadhimi, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020.

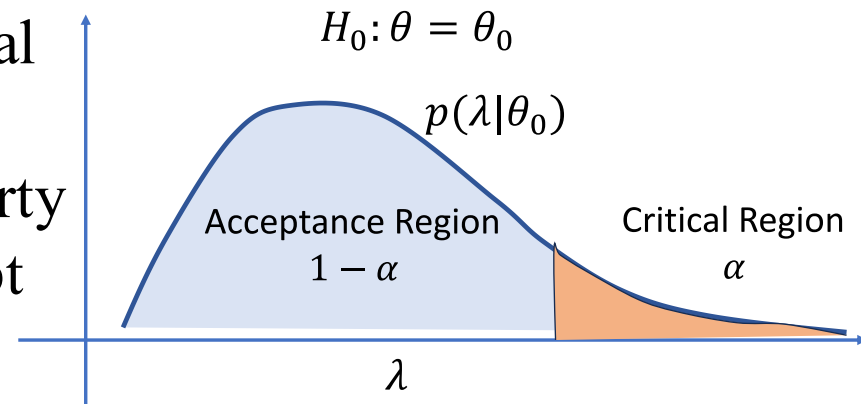
SIMULATION-BASED INFERENCE

SBI: Introduction

The key to SBI from the frequentist perspective is the relationship between hypothesis tests and confidence sets.

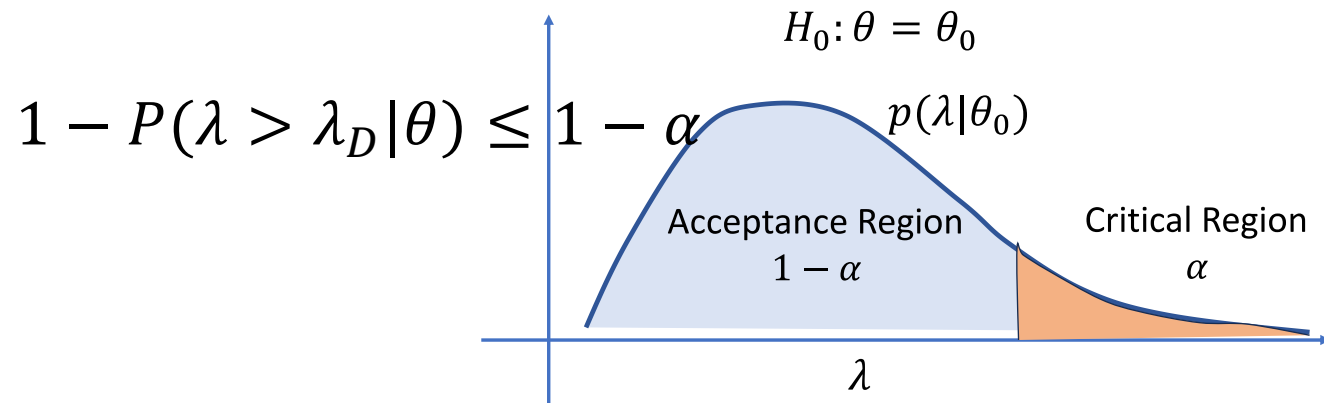
A hypothesis test $H_0: \theta = \theta_0$ typically proceeds as follows.

- Construct a function of potential observations X called a test statistic $\lambda(X, \theta)$ with the property that large values of λ cast doubt on the hypothesis H_0 .
- Choose a small *fixed* probability α .
- Compute $p = P(\lambda > \lambda_D | \theta)$, where $\lambda_D = \lambda(D, \theta)$.
- If $p < \alpha$, reject the hypothesis, otherwise do *not* reject it.



SBI: Introduction

- For a given dataset $X = D$, repeated application of this test for different values of $\theta = \theta_0$ divides the parameter space into those θ_0 that are *rejected* and those that are *not*.
- The set of θ_0 that are *not* rejected is called a **confidence set** with associated **confidence level $1 - \alpha$** .
- Consequently, the criterion for *not* rejecting the hypothesis $H_0: \theta = \theta_0$, and, therefore, adding θ_0 to the confidence set is:



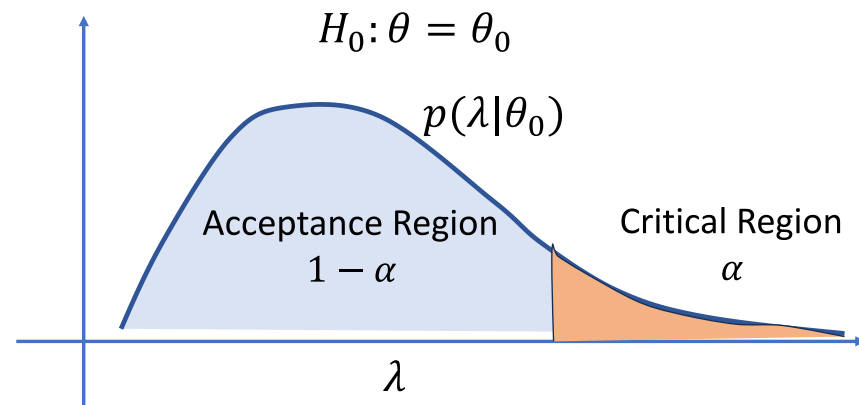
SBI: Introduction

- The key technical task is approximating either the *p-value*

$$p = P(\lambda > \lambda_D | \theta),$$

or, equivalently, the *cumulative density function* (cdf)

$$C(\lambda_D | \theta) \equiv 1 - p = P(\lambda \leq \lambda_D | \theta).$$



LIKELIHOOD FREE FREQUENTIST INFERENCE (LF2I)

LF2I

LF2I¹ is a suite of procedures for implementing frequentist-based inference when the likelihood function is intractable.

We'll focus on one aspect of LF2I, namely, the algorithm for approximating $P(\lambda \leq \lambda_D | \theta)$.

This algorithm requires:

1. A prior density $\pi(\theta)$ for sampling the parameter space.
2. A simulator $\mathcal{F}(\theta)$ to simulate potential observations, X .
3. A test statistic $\lambda(X, \theta)$.

1. Likelihood-Free Frequentist Inference: Confidence Sets with Correct Conditional Coverage Niccolo Dalmaso, Luca Masserano, David Zhao, Rafael Izbicki, Ann B. Lee, arXiv:2107.03920v6 [stat.ML] 6 Apr 2023.

LF2I: Algorithm

Algorithm 1 LF2I approximation of $\mathbb{C}(\lambda_D|\theta)$ given a simulator $\mathcal{F}(\theta)$

1. Initialize training sample $\mathbb{T} \leftarrow \emptyset$
- while** $k \in [1, \dots, K]$ **do**
 2. Sample $\theta_k \sim \pi(\theta)$
 3. Sample $X_k \equiv X_{1,k}, \dots, X_{n,k} \sim \mathcal{F}(\theta_k)$
 4. Compute test statistic $\lambda_k \leftarrow \lambda(X_k, \theta_k)$
 5. Compute test statistic $\lambda_{D,k} \leftarrow \lambda(D, \theta_k)$
 6. Compute indicator $Z_k \leftarrow \mathbb{I}(\lambda_k \leq \lambda_{D,k})$
 7. Update training sample $\mathbb{T} \leftarrow \mathbb{T} \cup \{(\theta_k, Z_k)\}$
- end while**
8. Use $\mathbb{T}$ to train a machine learning (ML) model, $f(\theta; \omega)$, to approximate $\mathbb{C}(\lambda_D|\theta)$, where θ are the inputs to $f(\theta; \omega)$, and ω are the ML model parameters.

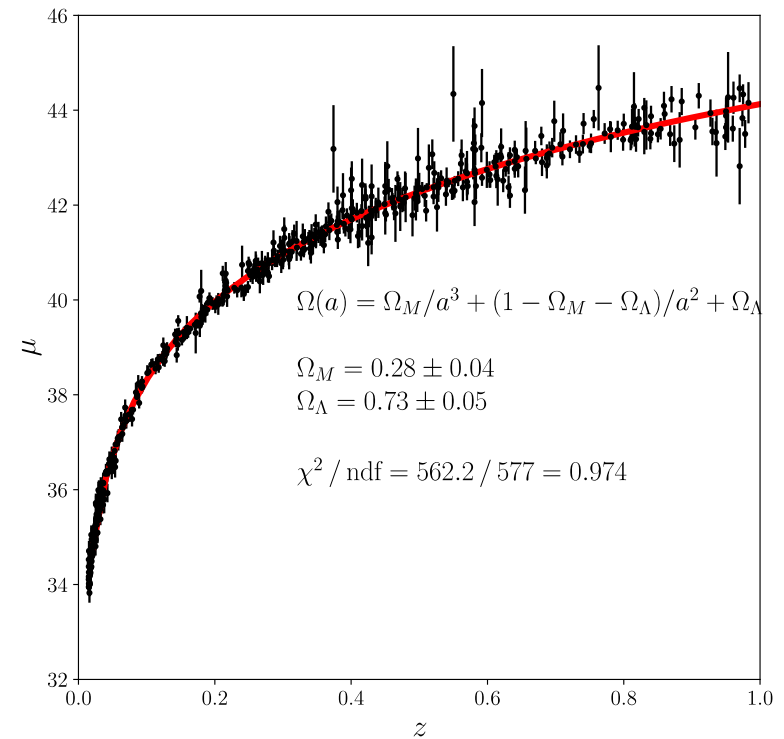
Neural network inputs: θ_k

Targets: Z_k

Loss function: $(Z_k - f)^2$ $f \rightarrow \mathbb{E}[Z | \theta] = P(\lambda \leq \lambda_D | \theta).$

LF2I: Example – Type 1a Sne

- Consider the problem of fitting a cosmological model, e.g., Λ CDM, to the [Union 2.1](https://www.supernova.lbl.gov/)¹ Type 1a supernova data set.
- The statistical model for this task is tractable (it is a 580-dimensional Gaussian), but the problem provides a simple benchmark to illustrate the LF2I algorithm, and a recent variation².



1. <https://www.supernova.lbl.gov/>

2. A. Al Kadhim, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020

LF2I: Example – Type 1a Sne

We'll use the test statistic

$$\lambda(X, \theta) = \sum_{i=1}^{580} \frac{(x_i - \mu(z_i, \theta))^2}{\sigma_i^2}$$

where $x_i \pm \sigma_i$ are the measured distance moduli, z_i the measured redshifts, θ the cosmological model parameters and $\mu(z_i, \theta)$ is the theoretical prediction for the distance modulus.

Our goal is to approximate $P(\lambda \leq \lambda_D | \theta)$.

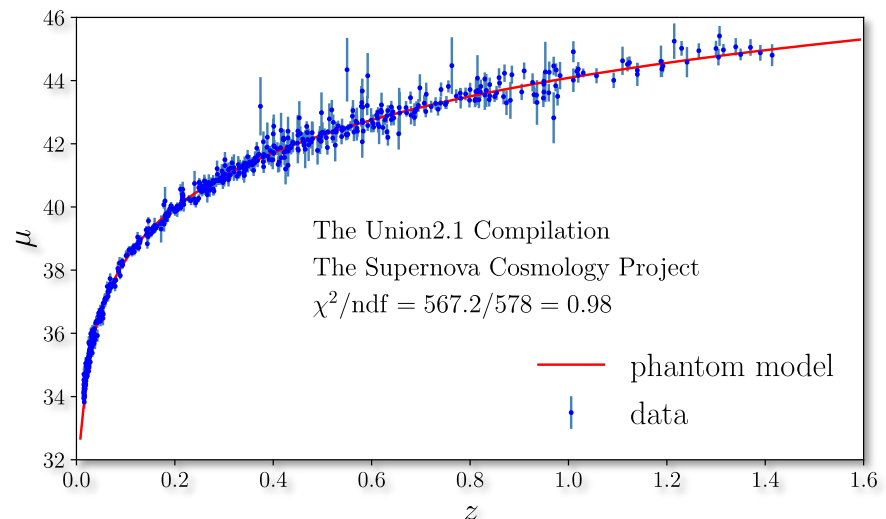
LF2I: Example – Type 1a Sne

We consider a cosmological model in which the energy density of the universe is given by

$$\Omega(a) = \frac{\exp(a^n - 1)}{a^3}$$

The model is defined by two parameters: the Hubble constant H_0 (not to be confused with the hypothesis symbol!) and the dimensionless parameter n .

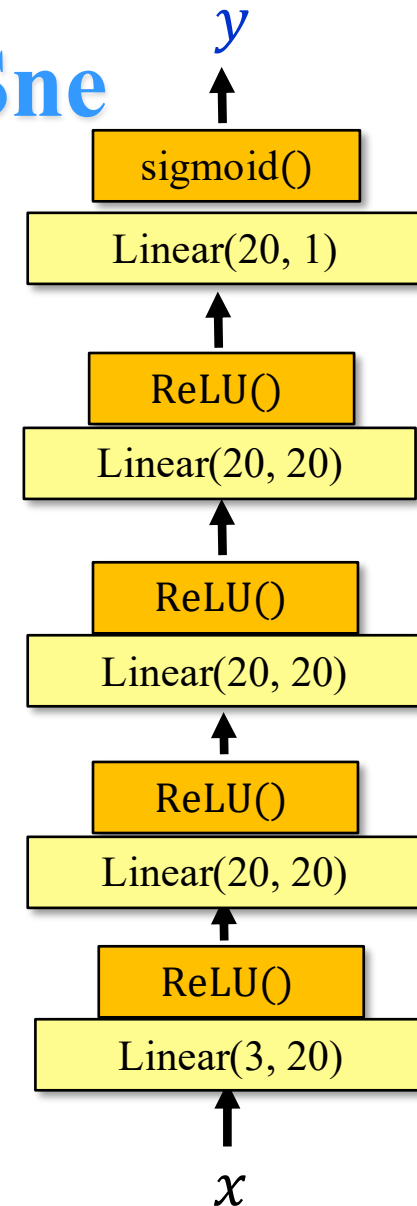
As the figure indicates, one gets an excellent fit of this model to the Type 1a data.



LF2I: Example – Type 1a Sne

Goal: model $y = P(\lambda \leq \lambda_D | \theta)$.¹

- Inputs: $x = \lambda_D, n, H_0$
- Training data: 250,000 sets of 580 distance moduli sampled from Gaussians with the same standard deviations as the real data.
- Batch size: 50.
- Learning rate: 10^{-3}
- Adam optimizer.
- 100,000 iterations.



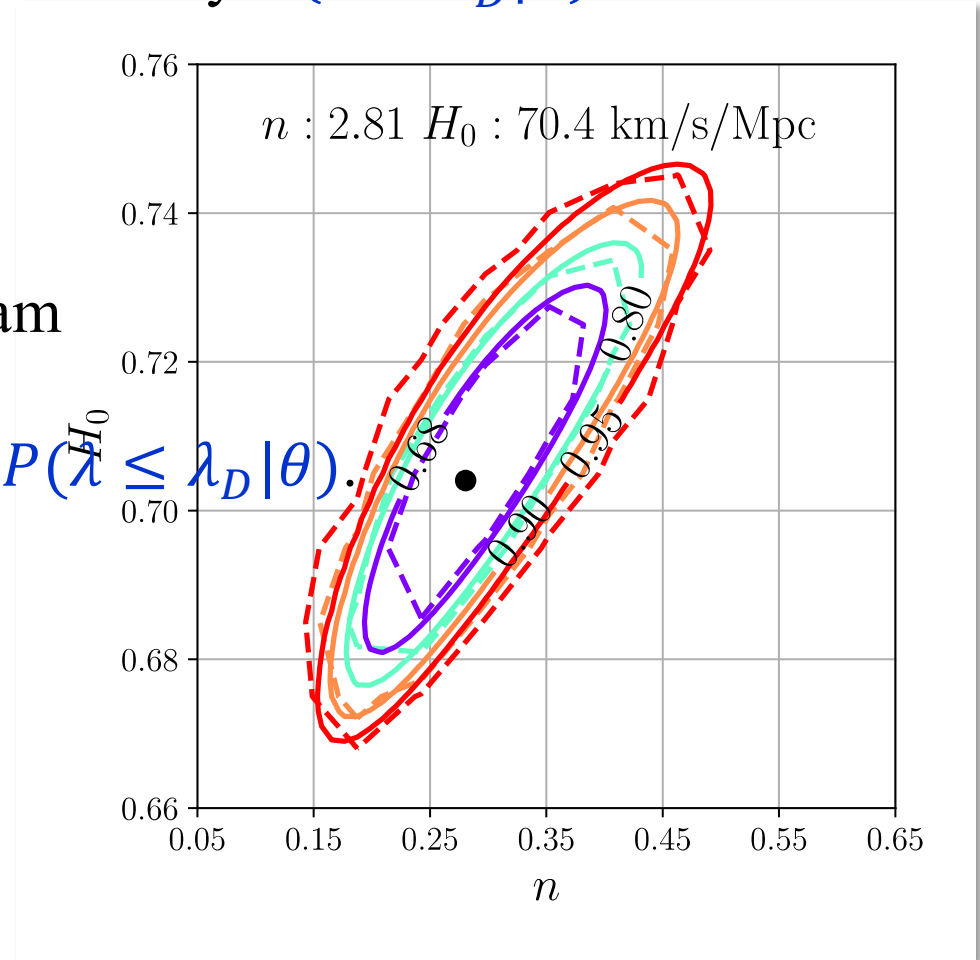
1. A. Al Kadhim, HBP, O. F. Prosper, *Mach. Learn.: Sci. Technol.* **5** (2024) 015020

LF2I: Example

Confidence set boundaries defined by $P(\lambda \leq \lambda_D | \theta) = 1 - \alpha$.

- Solid contours: NN.
- Dashed contours: histogram approximation of

$$E[Z | \theta] = P(\lambda \leq \lambda_D | \theta).$$



Summary

- Simulation-based inference is a set of methods to make inferences about the parameters of a statistical model when the latter is intractable.
- The key ingredients are:
 1. A prior density $\pi(\theta)$ for sampling the parameter space.
 2. A simulator $\mathcal{F}(\theta)$ to simulate potential observations, X .
 3. A test statistic $\lambda(X, \theta)$.
 4. Machine learning models to approximate various statistical quantities given data sets of real and simulated data.